
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): Zach McEwen

Student ID: 8768685

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Do not write in the table to the right.

Question	Points	Score
1	10	6
2	10	10
3	10	20
4	10	5
5	10	10+1
Total:	50	34

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$\iiint_T f(xu, yv, zw)$$

$$\iiint_T f(x(u,v,w), y(u,v,w), z(u,v,w)) \frac{\partial(x,y,z)}{\partial(u,v,w)} du dv dw$$

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \times \vec{F} \cdot \hat{n} dA$$

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

$$\text{Let } z - x^2 - y^2 = g$$

$$\vec{N} = \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k}$$

$$\vec{N} = \vec{\nabla} g = \begin{bmatrix} \partial g / \partial x \\ \partial g / \partial y \\ \partial g / \partial z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ 1 \end{bmatrix}$$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$\vec{F} = [x^2 + y^2, -2y]$$

Polar

$$\vec{F} = [r^2 \cos^2 \theta + r^2 \sin^2 \theta, -2r \sin \theta] = [r^2, -2r \sin \theta]$$

This is still using $\hat{i}, \hat{j}$
Need terms of $\hat{e}_r, \hat{e}_\theta$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$\nabla \cdot \vec{F} = r^2 + z^2$$

$$\nabla \cdot \vec{F} = \rho^2$$

$$\nabla \cdot \vec{F} = x^2 + y^2 + z^2$$

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cartesian

$$f(x, y, z) = x^2 + y^2 + z^2$$

cylindrical

$$f(r, \theta, z) = (r \cos \theta)^2 + (r \sin \theta)^2 + z^2 = r^2 + z^2$$

$$h_1 = 1 \quad h_2 = r \quad h_3 = 1$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(r \frac{\partial f}{\partial z} \right) \right]$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (r \cdot (2r \cos^2 \theta + 2r \sin^2 \theta)) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} (r \cdot 2z) \right]$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} 2r^2 + \frac{\partial}{\partial \theta} \frac{1}{r} \frac{\partial f}{\partial \theta} + 2r \right]$$

$$= \frac{1}{r} \left[4r + \frac{\partial}{\partial \theta} \right]$$

No θ goes to 0

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (r \frac{\partial}{\partial r} (r^2 + z^2)) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} (r \frac{\partial}{\partial z} (r^2 + z^2)) \right]$$

$$= \frac{1}{r} \left(\frac{\partial}{\partial r} r \cdot 2r + \frac{\partial}{\partial z} r \cdot 2z \right) = \frac{1}{r} (4r + 2r) = 6$$

Spherical
Back of formula sheet

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.

$$\phi = \frac{\pi}{6} = \text{const.}$$

$$0 \leq \rho \leq 2$$

$$x^2 + y^2 = z^2$$

$$= \int_0^{2\pi} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\theta$$

$$\rho^3 \, d\rho \, d\theta$$

$$dA = \rho^2 \sin \phi \, d\rho \, d\theta$$

$$x = \rho \cos \theta \sin \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \phi$$

 ρ, θ, ϕ

$$= \frac{1}{4} \pi \cdot \frac{\rho^4}{4} \Big|_0^2$$

$$\hat{n} = (2 \cos \theta \sin \phi, 2 \sin \theta \sin \phi, -2 \cos \phi)$$

$$= \frac{\pi}{4} \Big|_0^2 = 4\pi$$

$$= \rho^2 \cos^2 \theta \sin^2 \phi + \rho^2 \sin^2 \theta \sin^2 \phi - \rho^2 \cos^2 \phi$$

$$= \rho^2 (\cos^2 \theta \sin^2 \phi + \sin^2 \theta \sin^2 \phi - \cos^2 \phi)$$

$$R = \rho^2 (\sin^2 \phi - \cos^2 \phi) = -\rho^2$$

$$\vec{F} \cdot \hat{n} = \rho^3 \cos \phi = \rho^3$$

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

$$F = [x, y, z]$$

Solution.

$$\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\nabla \cdot F = 1 + 1 + 1 = 3$$

$$= \iiint 3 dV$$

$$= 3 \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^2 \sin\theta d\rho d\theta d\phi$$

$$= 3 \int_0^{2\pi} d\phi \int_0^{\pi/2} \sin\theta d\theta \int_0^2 \rho^2 d\rho$$

$$= 3 \cdot 2\pi \cdot (-\cos\theta) \Big|_0^{\pi/2} \cdot \frac{\rho^3}{3} \Big|_0^2$$

$$= -2\pi \cdot (0 - 1) \cdot 2^3 = 16\pi$$

$$dV = dx dy dz \Rightarrow \rho^2 \sin\theta d\rho d\theta d\phi$$

$$= \iint_S \vec{F} \cdot \hat{n} dA$$

$$= \int_0^{2\pi} \int_0^{\pi/2} 3 d\rho d\theta$$

$$= 3 \int_0^{2\pi} d\theta \int_0^2 d\rho = 12\pi$$

$$\vec{F} \cdot \hat{n} = 3 \quad x+y+z=3$$

$$dA = \rho^2 \sin\theta d\theta d\phi$$

$$= 4 \sin\theta d\theta d\phi \quad (\text{since } \rho=2)$$

$$\rightarrow = 3 \int_0^{2\pi} d\phi \int_0^{\pi/2} \int_0^2 \rho^2 d\rho$$

$$= 3 \cdot 2\pi \cdot \frac{\pi}{2} \cdot \frac{2^3}{3} = 8\pi^2$$

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Continue problem 4

Solution.

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$x = ve^u \quad y = ve^{-u}$$

$$\frac{dx}{du} = ve^u \quad \frac{dy}{du} = -ve^{-u}$$

$$\frac{dx}{dv} = e^u \quad \frac{dy}{dv} = e^{-u}$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} ve^u & e^u \\ -ve^{-u} & e^{-u} \end{vmatrix}$$

$$= ve^u \cdot e^{-u} + ve^{-u} \cdot e^u$$

$$= 2v \quad \checkmark$$

$$= \int_0^3 \int_0^u 2v \, dv \, du$$

$$= \int_0^3 v^2 \Big|_0^u \, du = \int_0^3 u^2 \, du = \frac{u^3}{3} \Big|_0^3 = 9 \quad \checkmark$$

Bonus +1 for
calculating
everything correctly

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$$h_1 = 1 \quad h_2 = \rho \sin \phi \quad h_3 = \rho$$

$$f(x, y, z) \Rightarrow f(\rho, \phi, \theta) = (\rho \cos \theta \sin \phi)^2 + (\rho \sin \theta \sin \phi)^2 + (\rho \cos \phi)^2$$

$$= \rho^2 (\cos^2 \theta \sin^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \phi)$$

$$= \rho^2$$

$$= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\rho^2 \sin \phi \frac{\partial}{\partial \rho} \rho^2 \right) + \frac{\partial}{\partial \phi} \left(\frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \rho^2 \right) + \frac{\partial}{\partial \theta} \left(\sin \phi \frac{\partial}{\partial \theta} \rho^2 \right) \right]$$

$$= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\rho^2 \sin \phi \cdot 2\rho \right) \right]$$

$$= \frac{1}{\rho^2 \sin \phi} \left[6\rho^2 \sin \phi \right] = 6$$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$| \text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$| \text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

$$\text{Laplacian of scalar field } f:$$

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$